Supplementary Material for "Cast-Break-Free Demand Response of Electric Arc Furnace Fleets: Exact Counting Polytopes and Certified Rolling Dispatch"

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This document gives the rows of the counting polytope (3) for the live state and the end of the horizon, the proofs of Theorems 1 to 4, and the example for Theorem 1(ii). Equations, theorems and Algorithm 1 are numbered as in the paper, and numbers with the prefix S belong to this document.

I. Boundary Rows of the Counting Polytope

Heats are indexed by start from the last starts of the live state, so heats 1 to n are the last heats of the n furnaces idle at the notice, which have tapped before it, and casts 1 to N^{C} are the last casts of the casters. First-in first-out casting gives the Q waiting ladles to casts $N^{\mathrm{C}}+1$ to $N^{\mathrm{C}}+Q$ and the ladle of heat $k > n$ to cast $k+q$, with $q = Q + N^{\mathrm{C}} - n$. The live state fixes the counters before $t = 0$. Row (3d) holds for $t \geq 0$ and row (3e) for the heats tapped from $t = 0$ on, and the j-th oldest waiting ladle, tapped at $\tau_j \leq 0$, adds the row $y(\tau_j + w) \geq N^{\mathrm{C}} + j$. Beyond H, rows (3b) and (3c) cap their arguments at H, and the other rows are dropped. Every row thus bounds the difference of two counters, or one counter, by an integer.

II. Proof of Theorem 1

For integer, non-decreasing counters, let

$$s_k = \min\{t : \ x(t) \geq k\}, \qquad z_k = \min\{t : \ y(t) \geq k\} \quad \text{(S1)}$$

be the start of heat k and the start of cast k, and let $v_k = s_k + d(s_k)$ be the tap of heat k. Then $x(t) \geq k$ if and only if $s_k \leq t$, and since $t + d(t)$ is non-decreasing, $v_k \leq t$ if and only if $x(\theta(t)) \geq k$.

Lemma S1. *For integer counters that satisfy* (3a)*, rows* (3b) *to* (3f) *hold at every slot if and only if, for every k whose starts lie in the horizon,*

$$s_{k+N^{\mathrm{E}}} \geq v_k + g, \quad \text{(S2a)}$$

$$z_k + c(z_k) \leq z_{k+N^{\mathrm{C}}} \leq z_k + c(z_k) + b, \quad \text{(S2b)}$$

$$v_k \leq z_{k+q} \leq v_k + w, \quad k > n. \quad \text{(S2c)}$$

Proof. Suppose (S2a) holds and fix t. The first heat started at or after t is $k = x(t-1)+1$, with $s_k \geq t$, and heat $k + N^{\mathrm{E}}$ starts at or after $v_k + g \geq t + d(t) + g$. At most N^{E} heats thus start in $[t,\, t + d(t) + g - 1]$, which is (3b). Conversely, (3b) at $t = s_k$ gives $x(v_k + g - 1) \leq x(s_k - 1) + N^{\mathrm{E}} \leq k - 1 + N^{\mathrm{E}}$, which is (S2a). The same argument with c, 0 and the casts shows that (3c) is equivalent to the left inequality of (S2b). If the right inequality holds and $k = y(t)$, cast $k + N^{\mathrm{C}}$ starts by $z_k + c(z_k) + b \leq t + c(t) + b$, which gives (3f), and (3f) at $t = z_k$ gives $y(z_k + c(z_k) + b) \geq k + N^{\mathrm{C}}$ back.

For (S2c), fix $t \geq 0$ and let $j = y(t)$. If $j \leq N^{\mathrm{C}} + Q$, then (3d) holds since $x(\theta(t)) \geq n$. Otherwise heat $j - q > n$ feeds cast j and taps by $z_j \leq t$, so $x(\theta(t)) \geq y(t) - q$. Conversely, (3d) at $t = z_{k+q}$ gives $x(\theta(z_{k+q})) \geq k$, so heat k taps by z_{k+q}. Finally, if $k = x(t) > n$, cast $k+q$ starts by $v_k + w \leq t + d(t) + w$, which gives (3e), and (3e) at $t = s_k$ gives $y(v_k + w) \geq k + q$, that is, $z_{k+q} \leq v_k + w$. □

Proof of Theorem 1. (i) Take a zero-break schedule and fix t. A furnace that starts a heat at $s \in [t,\, t + d(t) + g - 1]$ starts its next one at or after $s + d(s) + g \geq t + d(t) + g$ by (1d), so each furnace starts at most one heat in this window, which gives (3b). In the same way, (1e) lets each caster start at most one cast in $[t,\, t + c(t) - 1]$ and at least one in $(t,\, t + c(t) + b]$, which gives (3c) and (3f). The $y(t) - N^{\mathrm{C}}$ casts started by t after the last casts take distinct ladles among the Q waiting ones and the $x(\theta(t)) - n$ ladles tapped from $t = 0$ on by slot t, which gives (3d). By (1f), the waiting ladles and the ladles of the $x(t) - n$ heats $k > n$ started by t are cast by $t + d(t) + w$, which gives (3e) and the rows of the waiting ladles. Counters do not decrease after H, so capping arguments at H only weakens (3b) and (3c).

Conversely, let (x, y) be an integer point of $\mathcal{F}$. By Lemma S1, its starts satisfy (S2). Under the round-robin assignment, heat $k + N^{\mathrm{E}}$ is the next heat on the furnace of heat k, cast $k + N^{\mathrm{C}}$ is the next cast on the caster of cast k, and cast $k + q$ takes the ladle of heat k. Inequalities (S2) are therefore (1d) to (1f), and the rows of the waiting ladles give (1f) for the ladles waiting at the notice.

(ii) Each row of $\mathcal{F}$ has at most one coefficient $+1$ and one -1 and an integer right side, so the constraint matrix is the transpose of a network matrix, with unit rows added, and is totally unimodular [1]. The counters start at the live state and grow by at most N^{E} and N^{C} per slot by (3b) and (3c), so $\mathcal{F}$ is bounded and $\mathcal{F} = \mathrm{conv}(\mathcal{F} \cap \mathbb{Z})$. The block loads (1c) are linear in the counters, so they map $\mathcal{F}$ onto the convex hull of

the block loads of $\mathcal{F} \cap \mathbb{Z}$, which are those of the zero-break schedules by (i). □

III. Example for Theorem 1(ii)

Every edge of a g-polymatroid, and of its projections onto subsets of coordinates, is parallel to some e_j or $e_j - e_k$ [2]. Take one furnace and one caster with $d = 2$, $g = 0$, $c = 3$ and $b = w = 1$, an idle furnace, no waiting ladle and a cast in progress that ends at $t = 1$, with $t_0 = 0$ and $H \geq 10$. The first cast then starts at slot 2 with the ladle of a heat started at slot 0, and each later cast starts three or four slots after the previous one, with a ladle tapped at its start or one slot earlier. In units of $P/3$, the load vectors of the first three blocks are

$$(2,2,1),\ (2,2,2),\ (2,2,3),\ (2,3,1),\ (2,3,2),$$
$$(3,1,2),\ (3,1,3),\ (3,2,1),\ (3,2,2),$$

as the script `scripts/gpoly_example.py` of the repository of this document confirms. Heats started at slots 0, 3 and 8 give $(2,2,1)$, and heats started at slots 0, 2, 6 and 9 give $(3,1,2)$. The function $\ell_1 + 2\ell_2 + \ell_3$ equals 7 at these two vectors and at least 8 at the other seven, so the segment between them is an edge of the projected hull. Its direction $(1,-1,1)$ is parallel to no e_j and no $e_j - e_k$.

IV. Proof of Theorem 2

Proof. Step 1: arc costs. With $p_t = \underline{P} + (\overline{P} - \underline{P})\omega_t$ in (4),

$$\kappa(s,v) = \underline{P} \sum_{t=s}^{v-1} \pi_t + (\overline{P} - \underline{P})\, \phi\big(\{s,\dots,v-1\}, \nu_{v-s}\big), \quad \text{(S3)}$$

where $\phi(I,\nu)$ sums the ν cheapest prices of I, the last with a fractional weight, and $\nu_d = (E - d\underline{P})/(\overline{P} - \underline{P})$. The function $\phi(I,\cdot)$ is piecewise linear on $[0,|I|]$ with the sorted prices of I as slopes, hence convex, $\nu_{d+1} = \nu_d - \delta$ with $\delta = \underline{P}/(\overline{P} - \underline{P})$, and the first term of (S3) is modular in (s,v).

Step 2: local Monge inequality. Let $(s, v+1)$ and $(s+1, v)$ be admissible, so (s,v) and $(s+1, v+1)$ are too, and let $I = \{s+1,\dots,v-1\}$ and $\nu = \nu_{v-s}$. Let α and α' be the weights of slots s and v in $\phi(I \cup \{s,v\}, \nu - \delta)$, which then equals $\pi_s\alpha + \pi_v\alpha' + \phi(I, \nu - \delta - \alpha - \alpha')$. The numbers $\nu - \alpha$, $\nu - \alpha'$, $\nu - \delta - \alpha - \alpha'$ and $\nu + \delta$ lie in $[0,|I|]$, the smallest since $\alpha + \alpha' \leq \nu - \delta$ and the largest since $(s+1, v)$ is admissible. The same weights give

$$\phi(I \cup \{s\}, \nu) + \phi(I \cup \{v\}, \nu) \leq \pi_s\alpha + \pi_v\alpha' + \phi(I, \nu - \alpha) + \phi(I, \nu - \alpha'),$$

and the pairs $(\nu - \alpha, \nu - \alpha')$ and $(\nu - \delta - \alpha - \alpha', \nu + \delta)$ have the same sum, the second containing the first, so convexity bounds the right side by $\phi(I \cup \{s,v\}, \nu - \delta) + \phi(I, \nu + \delta)$. By (S3),

$$\kappa(s,v) + \kappa(s+1, v+1) \leq \kappa(s, v+1) + \kappa(s+1, v). \quad \text{(S4)}$$

Step 3: nested arcs. For $s_1 < s_2$, $v_1 < v_2$ and admissible nested arcs (s_1, v_2) and (s_2, v_1), all arcs in between are admissible, since admissible durations form an interval. Summing (S4) over $s_1 \leq s < s_2$ and $v_1 \leq v < v_2$ gives

$$\kappa(s_1, v_1) + \kappa(s_2, v_2) \leq \kappa(s_1, v_2) + \kappa(s_2, v_1). \quad \text{(S5)}$$

Step 4: uncrossing. Heats in progress only shift the integer right sides of the rows, so consider the new arcs. Moving the mass $\varepsilon = \min\{\xi(s_1,v_2), \xi(s_2,v_1)\}$ of a nested pair to (s_1,v_1) and (s_2,v_2) keeps x and $\hat{x}$, hence every row, does not raise the cost by (S5), and lowers $\sum \xi(s,v)(v-s)^2$ by $2\varepsilon(s_2 - s_1)(v_2 - v_1)$. An optimal solution that minimizes this sum among optimal solutions thus has no nested pair.

Step 5: rounding. Order the arcs of such a solution as $a_1,\dots,a_R$ with non-decreasing starts and taps, and let $M_j = \sum_{i \leq j} \xi(a_i)$. For $u \in [0,1)$, every difference constraint $a - a' \leq h$ with h integer gives

$$\lfloor a + u \rfloor - \lfloor a' + u \rfloor \leq \lceil a - a' \rceil \leq h, \quad \text{(S6)}$$

so the shifted floors of x, $\hat{x}$ and y keep every row [3]. The k-th rounded start and the k-th rounded tap both belong to the arc a_j with $j = \min\{j :\ M_j \geq k - u\}$, so every rounded heat is an arc of the support at its cheapest profile, and the rounded counters form a zero-break schedule with power within the band, as in Theorem 1. For u uniform, arc a_j carries $\xi(a_j)$ rounded heats on average and $\mathbb{E}\lfloor y(t) + u \rfloor = y(t)$, so the expected cost equals the LP value, and some u gives an integral optimal solution. □

V. Proof of Theorem 3

Proof of part (i). Step 1: feasibility. By (S6), every shifted floor (8) of a point of $\mathcal{F}_i$ lies in $\mathcal{F}_i \cap \mathbb{Z}$ [3], so by Theorem 1(i) it is a zero-break schedule.

Step 2: mean and range. For u uniform, $\lfloor a + u \rfloor$ is $\lceil a \rceil$ with probability $a - \lfloor a \rfloor$ and $\lfloor a \rfloor$ otherwise, so each rounded running count is $\lfloor r_i(t) \rfloor$ or $\lceil r_i(t) \rceil$ with mean $r_i(t)$. Hence $\mathbb{E}\,\ell_{i,\beta} = \ell^{\mathrm{LP}}_{i,\beta}$, and $\ell_{i,\beta}$ stays in an interval of length P_i.

Step 3: estimator. For each block, Markov's inequality applied to $e^{\lambda(\sum_i \ell_{i,\beta} - \bar{L}_\beta)}$ and the independence of the shifts bound the probability that the block reaches $C^{\mathrm{LP}} + \Delta$ by the term of β in (9), and the union bound gives $U(\Delta,\lambda)$ for the peak.

Step 4: derandomization. Let U_j be U with the shifts of the first j plants of $\mathcal{S}$ fixed. Since U_j is the average of U_{j+1} over the next shift, the minimizing shift gives $U_{j+1} \leq U_j$. With all shifts fixed, $U = \sum_\beta \exp\big(\lambda(\sum_i \ell_{i,\beta} - \bar{L}_\beta - C^{\mathrm{LP}} - \Delta)\big) < 1$, so every block lies below $C^{\mathrm{LP}} + \Delta$. The shifted floors of plant i change only where u_i crosses one minus the fractional part of one of the $O(m)$ counters in its block loads, so each expectation in (9) has $O(m)$ outcomes, and the rounding takes polynomial time.

Step 5: fractional plants. Let G_i be the face of $\mathcal{F}_i$ that contains (x_i, y_i) of an optimal vertex in its relative interior. The face $G = \prod_i G_i \times \mathbb{R}$ contains the vertex in its relative interior, and the vertex is the only nearby point of G at which the tight rows (1b) hold with equality, so $\dim G = 1 + \sum_i \dim G_i$ is at most their number, at most m. Since $\mathcal{F}_i$ has integral vertices, $\dim G_i \geq 1$ for $i \in \mathcal{S}$, so $|\mathcal{S}| \leq m - 1$, as in [4].

Step 6: certificate. Hoeffding's lemma gives $\mathbb{E}\, e^{\lambda(\ell_{i,\beta} - \ell^{\mathrm{LP}}_{i,\beta})} \leq e^{\lambda^2 P_i^2/8}$, and $C^{\mathrm{LP}}_\beta \leq C^{\mathrm{LP}}$, so with $\lambda = 4\Delta / \sum_{i \in \mathcal{S}} P_i^2$,

$$U(\Delta,\lambda) \leq m \exp\Big(-2\Delta^2 \Big/ \sum_{i \in \mathcal{S}} P_i^2 \Big), \quad \text{(S7)}$$

which is below one when $\Delta^2 > \frac{1}{2}\ln m \sum_{i\in\mathcal{S}} P_i^2$. By Step 5, $\sum_{i\in\mathcal{S}} P_i^2 \le (m-1)P_{\max}^2$, so every $\Delta > \left(\frac{1}{2}(m-1)\ln m\right)^{1/2} P_{\max}$ is a certificate. □

Proof of part (ii). Step 1: instance. Take sets $T_1, \dots, T_M$ on N elements, and give set k the blocks A_k of slots $6k$ to $6k+2$ and B_k of slots $6k+3$ to $6k+5$, so $m = 2M$. Element i becomes plant i, with one furnace of power $P = P_{\max}$, one caster, $g = w = 0$, $b = 1$ and heat times $3, 4, 7, 6, 5, 4$ for start slots 0 to 5 modulo six. Then $s + d(s)$ is non-decreasing, and taps fall only on slots 3 and 5 modulo six. The casting times make a cast started at slot $6k+3$ or $6k+5$, with $i \in T_k$, last until the same slot of the next set that contains i, and a lead-in before the window lets the first cast of plant i start at slot 3 or 5 of its first set.

Step 2: two schedules. Since $w = 0$ and $b = 1$, every cast starts at a tap at the end slot of the previous cast, so the offset, 3 or 5, is the same in all sets of plant i. Let $\chi_i = +1$ for offset 3 and $\chi_i = -1$ for offset 5. A tap at $6k+5$ needs the heat started at $6k+1$, which runs in slots $6k+1$ to $6k+4$, and a tap at $6k+3$ needs the heat started at $6k$, which runs in slots $6k$ to $6k+2$, or a longer heat started earlier. The two minimal schedules thus add $P(5+\chi_i)/6$ to A_k and $P(1-\chi_i)/3$ to B_k for every set k that contains i, and every schedule of plant i is dominated block by block by the minimal one of its sign.

Step 3: gap. With the baselines $\bar{L}_{A_k} = 5P|T_k|/6 - C_0$ and $\bar{L}_{B_k} = P|T_k|/3 - C_0$, the minimal schedules give the excesses $C_0 + P\sigma_k/6$ in A_k and $C_0 - P\sigma_k/3$ in B_k, with $\sigma_k = \sum_{i\in T_k} \chi_i$. By Theorem 1(ii), the fleet LP may take $\chi \in [-1,1]^N$, and it attains C_0 at $\chi = 0$ and no less, since one of the two excesses is at least C_0. Hence

$$C^\star - C^{\mathrm{LP}} = \min_{\chi\in\{\pm1\}^N} \max_k \max \left\{ \frac{P}{6}\sigma_k, -\frac{P}{3}\sigma_k \right\}, \tag{S8}$$

which lies between $P/6$ and $P/3$ times the discrepancy $\mathrm{disc}(T) = \min_\chi \max_k |\sigma_k|$.

Step 4: growth. For a Hadamard matrix H of order N, a power of two, whose first row has all entries $+1$, let $T_k = \{i : H_{ki} = 1\}$. For $\chi \in \{\pm1\}^N$ and $\Sigma = \sum_i \chi_i$, we have $\sigma_1 = \Sigma$ and $(H\chi)_k = 2\sigma_k - \Sigma$. Since $\|H\chi\|^2 = N^2$, some k has $|2\sigma_k - \Sigma| \ge \sqrt{N}$, so $\max_k |\sigma_k| \ge \max\{|\Sigma|, (\sqrt{N} - |\Sigma|)/2\} \ge \sqrt{N}/3$. With $M = N$, (S8) gives $C^\star - C^{\mathrm{LP}} \ge P\sqrt{N}/18 = \Omega(\sqrt{m}\, P_{\max})$.

Step 5: hardness. For set systems with $\Theta(N)$ sets on N elements, it is NP-hard to tell discrepancy zero from discrepancy at least $c\sqrt{N}$, for some constant $c > 0$ [5]. The instance has polynomial size and $m = \Theta(N)$, and by (S8), $C^\star = C^{\mathrm{LP}}$ in the first case and $C^\star \ge C^{\mathrm{LP}} + cP\sqrt{N}/6$ in the second. A polynomial algorithm that guaranteed a peak within $\epsilon\sqrt{m}\,P$ of $C^\star$, with $\epsilon\sqrt{m} < c\sqrt{N}/6$, would tell the two cases apart by comparing its peak with $C^{\mathrm{LP}} + cP\sqrt{N}/6$. □

VI. Proof of Theorem 4

Lemma S2. *Let s_k be the k-th heat start. For all heat times within the bounds that agree with the observations up to slot t, the k-th tap falls in $[\underline{\eta}_t(s_k), \bar{\eta}_t(s_k)]$.*

Proof. The heats with start index $k' \le k$ start at $s_{k'} \le s_k$ and tap by $s_{k'} + \overline{d}_t(s_{k'}) \le \bar{\eta}_t(s_k)$, so at least k taps occur by $\bar{\eta}_t(s_k)$. The heats with $k' \ge k$ tap at or after $s_{k'} + \underline{d}_t(s_{k'}) \ge \underline{\eta}_t(s_k)$, so at most $k-1$ taps occur before $\underline{\eta}_t(s_k)$. □

Proof of Theorem 4. (i) *Step 1: every integer point of $\mathcal{R}_t$ is safe.* Let p be an integer point of $\mathcal{R}_t$, executed with the round-robin assignment and first-in first-out ladles, and fix heat times within the bounds that agree with the observations up to t. Row (11a) at $s = s_k$ gives $x(\bar{\eta}_t(s_k) + g - 1) \le k - 1 + N^{\mathrm{E}}$, so heat $k + N^{\mathrm{E}}$, the next heat on the furnace of heat k, starts at least g slots after $\bar{\eta}_t(s_k)$, which is at least the latest tap $s_k + \overline{d}_t(s_k)$ of heat k. If cast $k+q$ starts at z, (11b) at $s = z$ gives $x(\bar{\theta}_t(z)) \ge k$, so $\bar{\eta}_t(s_k) \le z$, and (11c) at $s = s_k$ gives $z \le \underline{\eta}_t(s_k) + w$. By Lemma S2, the k-th tapped ladle, which cast $\overline{k} + q$ receives, is ready by z and waits at most w. Rows (3c) and (3f) involve no heat times, so Lemma S1 gives (1e). The schedule of p thus satisfies (1d) to (1f) at these heat times.

Step 2: recursive feasibility. Let p_t be the point of $\mathcal{R}_t$ whose slot-t starts are executed. As the residual intervals shrink, $\bar{\eta}_{t+1} \le \bar{\eta}_t$, $\underline{\eta}_{t+1} \ge \underline{\eta}_t$ and $\bar{\theta}_{t+1} \ge \bar{\theta}_t$, so since the counters are non-decreasing, every row of $\mathcal{R}_{t+1}$ holds at p_t, whose starts before $t+1$ are the executed ones. Hence $p_t \in \mathcal{R}_{t+1}$, and by induction every $\mathcal{R}_{i,t}$ is feasible.

Step 3: conclusion. At the last slot T, the executed schedule coincides with p_T, and the realized heat times agree with all observations, so by Step 1 the executed schedule is zero-break at the realized heat times, for all heat times within the bounds.

(ii) The rows of $\mathcal{R}_{i,t}$ are difference constraints with integer data, so by (S6) every combination of shifted floors that Algorithm 1 may execute lies in $\mathcal{R}_{i,t}$. Algorithm 1 thus respects every $\mathcal{R}_{i,t}$, and (i) applies. At an optimal vertex of (12), every left row of (12b) is tight, since a smaller C_β would lower (12a), and let $\mu \le m_t$ right rows be tight. The face count of Step 5 of part (i) of Theorem 3, now with the $1 + m_t$ variables C and C_β, gives $1 + m_t + \sum_i \dim G_i \le m_t + \mu$, so at most $m_t - 1$ plants are fractional, and Steps 1 to 4 and 6 hold for the m_t remaining blocks with the peak C of the vertex in place of C^{LP}. The search accepts only a lower planned peak, so it keeps the guarantee. □

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